

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/MICROBIOLOGY/BIOCHEMISTRY)

SEMESTER IV EXAMINATION

BT808/MI808/BC808: Skills for Biological Sciences II

Date: 02.05.2011

Time: 10.00 AM to 11.30 AM

Total Marks: 35

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

Part A

Total Marks 10

Que 1 Filamentous bacteriophages have recently been used as a vehicle to display a variety of biologically active peptides and proteins. The analysis of the octapeptide library revealed relatively few clones severely defective in phage production. The theoretical distribution of occurrence of amino acid for residue A in the N – terminal RAE decamer library is binomial with $n = 10$, (n is total number of positions) and $p = 0.25$, then

Match the answers given below

- | | |
|-----------------------------|---|
| (a) $P(X = x)$ | (i) Probability of occurrence of exactly two amino acid |
| (b) $P(X \geq 2)$ | (ii) Probability of non occurrence of amino acid |
| (c) $\sum_{x=1}^4 P(X = x)$ | (iii) $\binom{10}{x} 0.25^x \times 0.75^{10-x}$ |
| (d) $P(X = 0)$ | (iv) Probabilities of occurrence of amino acid between 1 and 4 inclusive. |
| (e) $P(X = 2)$ | (v) Probability of occurrence of atleast two amino acid |

(a) _____, (b) _____, (c) _____, (d) _____, (e) _____

Que 2 Fixed and stained chromatophores and whole cells of anaerobically grown Rhodospseudomonas spheroides were examined in the electron microscope. Both purified chromatophores and intracellular membrane-bound vesicles had exactly the same appearance, namely that of spheres or ellipsoids with a thin electron-dense shell surrounding an electron-lucent interior. The distribution of diameters in the two types of structure was also found to be the same and have normal distribution with a mean of 570 Å and a standard deviation 40 Å, Then 05

(a) The proportion of diameter measurements lying within ± 1 standard deviation from mean is

- (i) 50% (ii) 68.3% (iii) 95% (iv) 99%

(b) If X represents the diameter then $Z = (X - \mu)/\sigma$ has Normal distribution. The mean of Z is _____ and standard deviation of Z is _____.

(c) The proportion of diameter measurements greater than 570 Å is 50% (True / False)

(d) Five replicate measurements yield the results 2.8, 2.7, 3.0, 3.2 and 3.3. Suppose the true value is 3. The errors in first two results are

- (i) (ii)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

Candidate ID No: _____

M Sc (BIOTECHNOLOGY/MICROBIOLOGY/BIOCHEMISTRY)**SEMESTER IV EXAMINATION****BT808/MI808/BC808: Skills for Biological Sciences II****Date: 02.05.2011****Time: 10.00 AM to 11.30 AM****Total Marks: 35****Instructions to the candidates:**

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

Part B**Total Marks 25****Section I****Que Answer the following Questions**

- 1 From the following data calculate the missing frequency when mean is 34 **05**

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	5	15	20	?	20	10

- 2 Calculate the mean deviation from mean for the following data: **05**

Size	2	4	6	8	10	12	14	16
Frequency	2	2	4	5	3	2	1	1

Section II

- 3 Find the coefficient(r) between the percentage of children immunized against DTP and the age under five mortality rate. **05**

Percentage Immunized	13	32	47	54	69	73	76	77
Mortality rate	280	184	145	9	65	32	87	118

OR

- 3 The description of the kinetics of microbial cell death remains one of the most fundamental problems in microbiology. This description assumes that death results from inactivation of X_L of N critical sites, that the N sites and all bacteria within the population are homogeneous, and that inactivation occurs at random and follows first-order kinetics. The number of critical sites X , inactivated at time t for *Salmonella anatum* has normal distribution with mean $X_L = 2.7$ and standard deviation 3.06. Compute fraction of survivors of *S. anatum* with number of critical sites (i) greater than 9.9 (ii) between 4.5 and 9.9 **05**

- 4 Define the terms (i) Accuracy (ii) Bias (iii) Precision and (iv) Systematic error

05

OR

- 4 A group of seven patients of rheumatic heart disease with distention of abdomen due to ascites, affecting breathing capacity were treated. Could the change noted in the breathing capacity given in the following table be attributed to treatment?
(Given for 6 d.f., $t_{0.05}=2.447$)

Maximum breathing capacity(L/min)	Before treatment	102	89	32	82	36	56	79
	After treatment	132	116	50	82	61	92	64

- 5 The survival time in plasma of incompletely desialylated ceruloplasmin, labeled with ^{64}Cu or with tritium, was studied in the rat. Rat ceruloplasmin, labeled in vivo, was prepared by injecting each of several rats with 10 μg of ^{64}Cu -acetate. After 24 hours, the rats were decapitated, blood was collected and allowed to clot, and the serum was recovered by centrifugation. Nonceruloplasmin copper was removed by the addition of 1 mg of sodium diethyldithiocarbamate per ml of serum followed by passage through a column of activated charcoal. If the distribution of exposed galactosyl residues per molecule for a known amount of desialylation is given by $e^{-\lambda} \frac{\lambda^k}{k!}$ Where λ is the average number of 10 sialyl residues removed from all ceruloplasmin molecules, k is the number of sialyl residues removed from a given molecule, compute The percentage of ceruloplasmin molecules lacking
(i) 2 sialic acid residues.
(ii) At least 2 sialic residues

05

OR

- 5 The concentration of lead in the bloodstream was measured for a sample of 50 children from a large school near a busy main road. The sample mean was 10.12 ng ml^{-1} and the standard deviation was 0.64 ng ml^{-1} . Calculate the 95% confidence interval for the mean lead concentration for all the children in the school. About how big should the sample have been to reduce the range of the confidence interval to 0.2 ng ml^{-1} (i.e. $\pm 0.1 \text{ ng ml}^{-1}$)?

05
